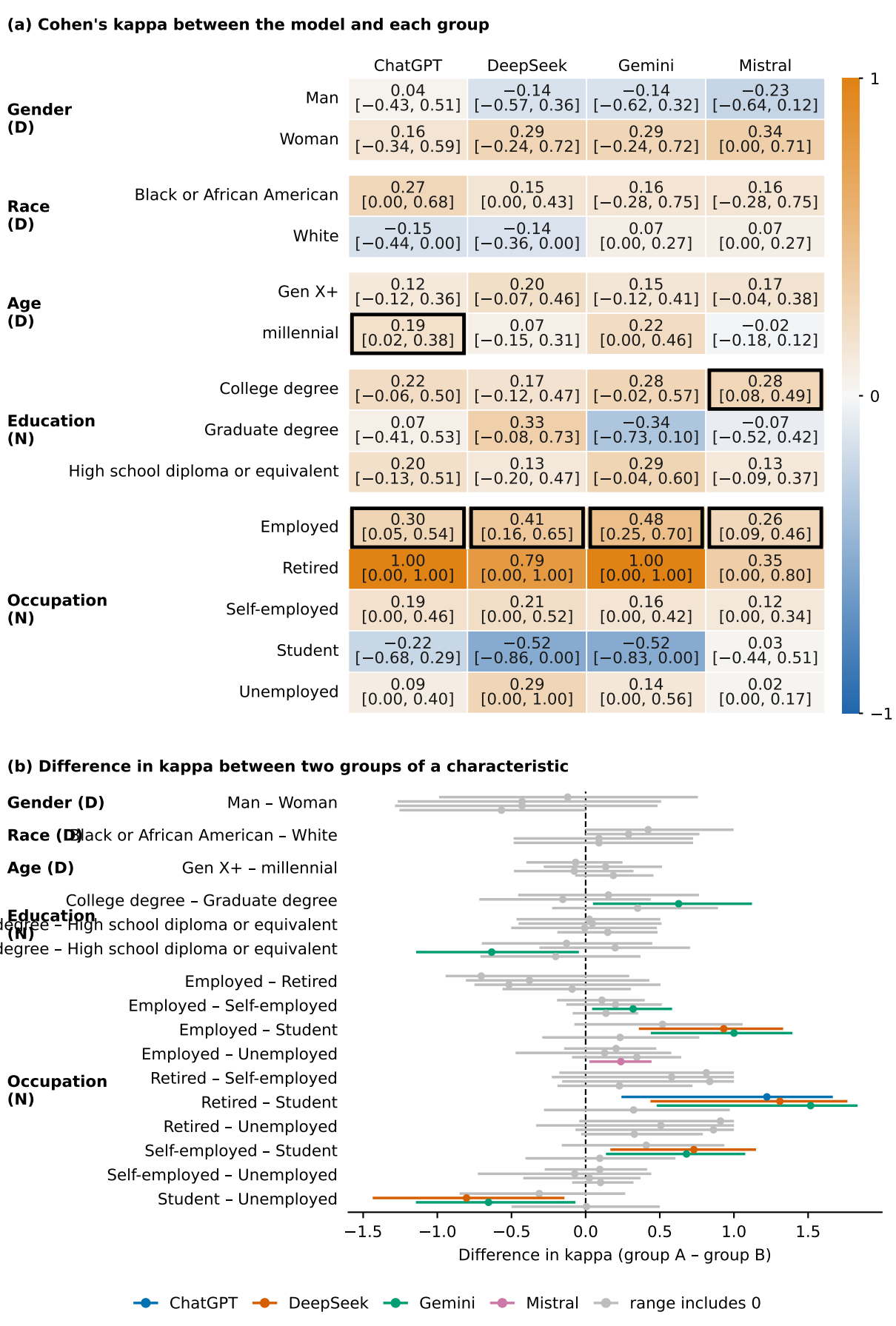
(a) Cohen's kappa between the model and each group
ChatGPT
DeepSeek
Gemini
Mistral
Gender (D)
Man
0.04 [−0.43, 0.51]
−0.14 [−0.57, 0.36]
−0.14 [−0.62, 0.32]
−0.23 [−0.64, 0.12]
Woman
0.16 [−0.34, 0.59]
0.29 [−0.24, 0.72]
0.29 [−0.24, 0.72]
0.34 [0.00, 0.71]
Race (D)
Black or African American
0.27 [0.00, 0.68]
0.15 [0.00, 0.43]
0.16 [−0.28, 0.75]
0.16 [−0.28, 0.75]
White
−0.15 [−0.44, 0.00]
−0.14 [−0.36, 0.00]
0.07 [0.00, 0.27]
0.07 [0.00, 0.27]
Age (D)
Gen X+
0.12 [−0.12, 0.36]
0.20 [−0.07, 0.46]
0.15 [−0.12, 0.41]
0.17 [−0.04, 0.38]
millennial
0.19 [0.02, 0.38]
0.07 [−0.15, 0.31]
0.22 [0.00, 0.46]
−0.02 [−0.18, 0.12]
Education (N)
College degree
0.22 [−0.06, 0.50]
0.17 [−0.12, 0.47]
0.28 [−0.02, 0.57]
0.28 [0.08, 0.49]
Graduate degree
0.07 [−0.41, 0.53]
0.33 [−0.08, 0.73]
−0.34 [−0.73, 0.10]
−0.07 [−0.52, 0.42]
High school diploma or equivalent
0.20 [−0.13, 0.51]
0.13 [−0.20, 0.47]
0.29 [−0.04, 0.60]
0.13 [−0.09, 0.37]
Occupation (N)
Employed
0.30 [0.05, 0.54]
0.41 [0.16, 0.65]
0.48 [0.25, 0.70]
0.26 [0.09, 0.46]
Retired
1.00 [0.00, 1.00]
0.79 [0.00, 1.00]
1.00 [0.00, 1.00]
0.35 [0.00, 0.80]
Self-employed
0.19 [0.00, 0.46]
0.21 [0.00, 0.52]
0.16 [0.00, 0.42]
0.12 [0.00, 0.34]
Student
−0.22 [−0.68, 0.29]
−0.52 [−0.86, 0.00]
−0.52 [−0.83, 0.00]
0.03 [−0.44, 0.51]
Unemployed
0.09 [0.00, 0.40]
0.29 [0.00, 1.00]
0.14 [0.00, 0.56]
0.02 [0.00, 0.17]
1
0
−1
(b) Difference in kappa between two groups of a characteristic
Gender (D)
Man – Woman
Race (D)
Black or African American – White
Age (D)
Gen X+ – millennial
Education (N)
College degree – Graduate degree
degree – High school diploma or equivalent
degree – High school diploma or equivalent
Occupation (N)
Employed – Retired
Employed – Self-employed
Employed – Student
Employed – Unemployed
Retired – Self-employed
Retired – Student
Retired – Unemployed
Self-employed – Student
Self-employed – Unemployed
Student – Unemployed
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Difference in kappa (group A – group B)
ChatGPT
DeepSeek
Gemini
Mistral
range includes 0